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5. (*a*) What are the different types of power transistors? 2

(*b*) Explain the constructional details of PMOSFET and describe its switching characteristics. 7

(*c*) What is an IGBT? Give its basic structural features and derive the approximate and exact equivalent circuits. 7

(*d*) A separately-excited dc motor, operating from a single-phase half-contolled bridge at a speed of 1400 rpm, has an input voltage of 330 sin 314t and a back emf 80 V. The SCRs are fired symmetrically at $\alpha = 30°$ in every half cycle and the armature has a resistance of 4Ω. Calculate the average armature current and the motor torque. 7

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324613(24)

BE (6th Semester)
Examination, April - May, 2016

(Old Scheme)

Power Electronics

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Time Allowed : 3 hours *Maximum Marks* : 80
Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory. Attempt any **two** parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) Define latching current and holding current. 2

(*b*) Draw and explain the circuit model of a triggering circuit connected to the gate-cathode terminals of a Thyristor. Explain the purpose of connecting a resistor across the gate circuit of an SCR. 7

(*c*) Explain the different mechanisms by which Thyristors can be triggered into conduction. 7

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(*d*) Gate-cathode characteristics of a Thyristor have a spread given by the following two relations. 7

$I_g = 2.1\times10^{-3}\ V_g^{\ 2}$ and $I_g = 2.1\times10^{-3}\ V_g 1.5$

The gate source voltage is 16 V and load line has a slope of 128 V/A. Calculate the trigger voltage and trigger current for an average gate power dissipation of 0.5 W.

2. (*a*) Draw the circuit diagram for single-phase full wave Mid-point converter. 2

(*b*) A single-phase transformer, with secondary voltage of 230 V, 50 Hz, delivers power to the load R = 10 Ω through a half-wave controlled rectifier circuit. For a firing-angle delay of 60°, determine (i) the rectification efficiency (ii) form factor (iii) voltage ripple factor (iv) Transformer utilization factor and (v) PIV of Thyristor. 7

(*c*) Discuss the effect of source inductance on the performance of a single-phase full converter indicating clearly the conduction of various Thyristors during one cycle. 7

(*d*) A three-phase M-3 converter is operated from 3-phase, 230 V, 50 Hz supply with load resistance R = 10Ω. An average output voltage of 50% of the maximum possible output voltage is required. Determine (1) the firing angle (2) average and rms values of load current and (3) rectification efficiency. 7

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3. (*a*) What is time ratio control in DC Choppers? 2

(*b*) Describe the principle of step-up chopper. Derive an expression for the average output voltage in terms of input voltage and duty cycle. 7

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(*c*) Explain the different methods of commutation in chopper. 7

(*d*) A voltage commutated chopper feeds power to a battery-powered electric car. The battery voltage is 60 V, starting current is 60 A and Thyristor turn off time is 20μsec. Calculate the values of the commutating capacitor C and the commutating inductor L. 7

4. (*a*) What are line commutated inverters? 2

(*b*) Explain the operating principle of single-phase to single-phase setp up cycloconverter with the help of mid-point and bridge type configurations. 7

(*c*) Describe the working of a single-phase parallel inverter with relevant circuit and waveforms. 7

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(*d*) Calculate the output frequency of a series inverter with the following parameters; Inductance = 6 mH, capacitance = 1.2μF, load resistance R = 100Ω, T_{off} - 0.2ms. If the load resistance is varied from 40 to 140 ohms, find out the range of output frequency. 7